

Clinical Risk Assessment Report: Patient 22

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.47x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.5%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.18 [Avg (39th %ile)] (-2.5%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.64 [Bottom 19%] (-1.1%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.69 [Bottom 13%] (-0.9%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -1.1 [Bottom 6%] (-0.8%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.25 [Top 5%] (-0.8%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.62 [Top 4%] (-0.8%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.73 [Bottom 22%] (-0.3%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.51 [Top 5%] (-0.2%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

DISCLAIMER: This report is generated by an investigational machine learning model. It is intended to provide adjunctive biological insights and must not replace standard clinical, radiological, or pathological evaluation.

Machine Learning Feature Attribution (SHAP Decision Path)

